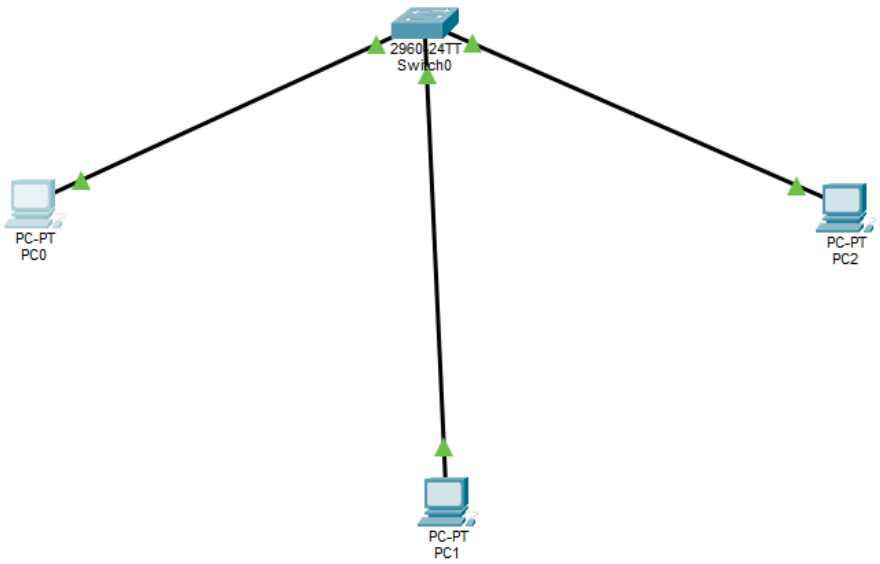
Computer Network Lab-3

Slot: L3+L4

Aswini Ambili, 24BCE0776

1. Three computers have been connected in the same office LAN through a switch using star. The administrator claims that all devices have been configured correctly.
   a. Confirm whether every host can communicate with the remaining hosts.
   b. Determine whether the operating system has learned the physical addresses of neighboring devices.
   c. Identify the forwarding information learned by the switch after communication.



1.a

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

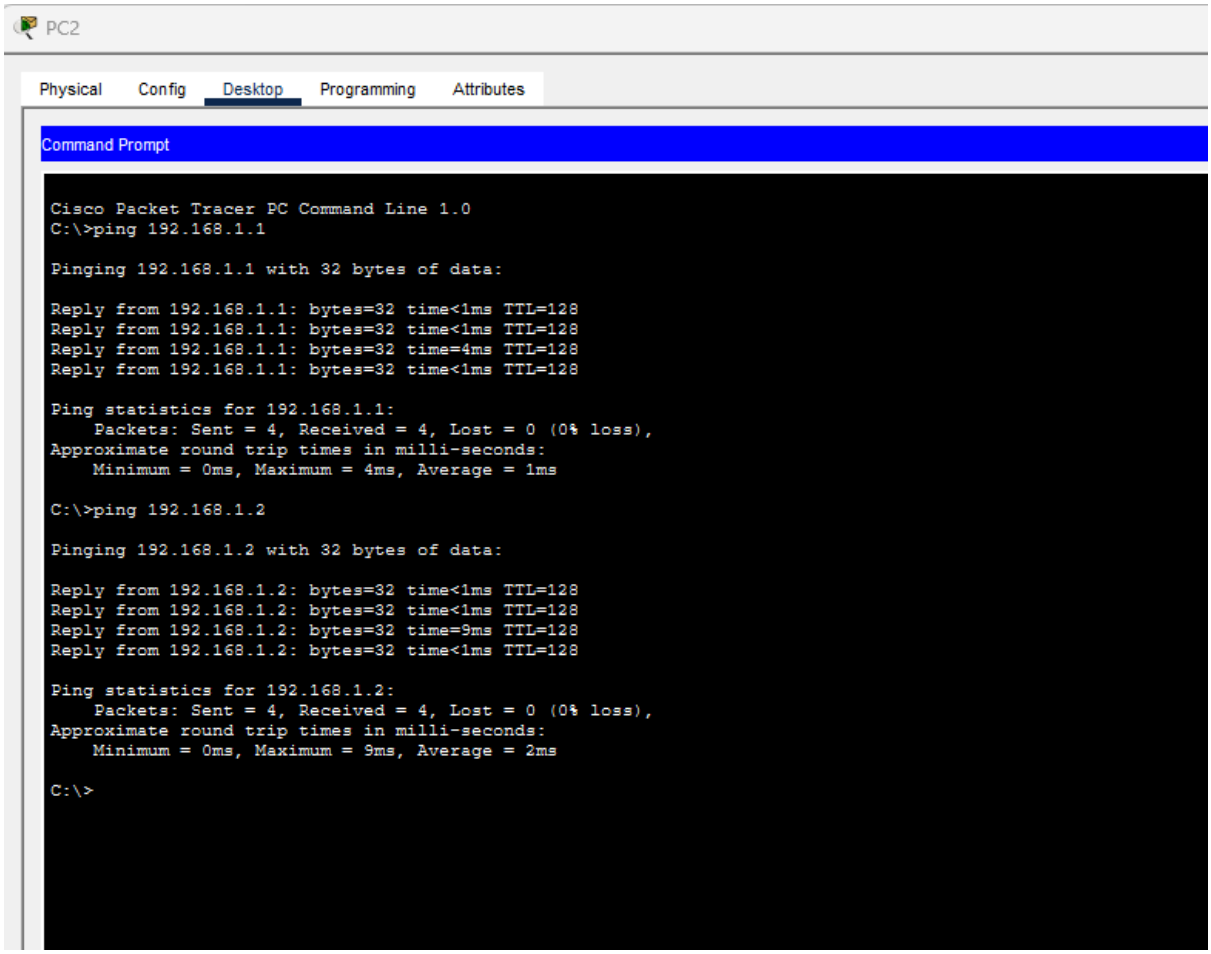
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```



1.b

PC0

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.2           0003.e470.e4e6        dynamic
  192.168.1.3           00e0.a3de.0ae3        dynamic
```

PC1

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0060.3e7e.de13        dynamic
  192.168.1.3           00e0.a3de.0ae3        dynamic
```

PC2

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0060.3e7e.de13        dynamic
  192.168.1.2           00e0.f778.4888        dynamic
```

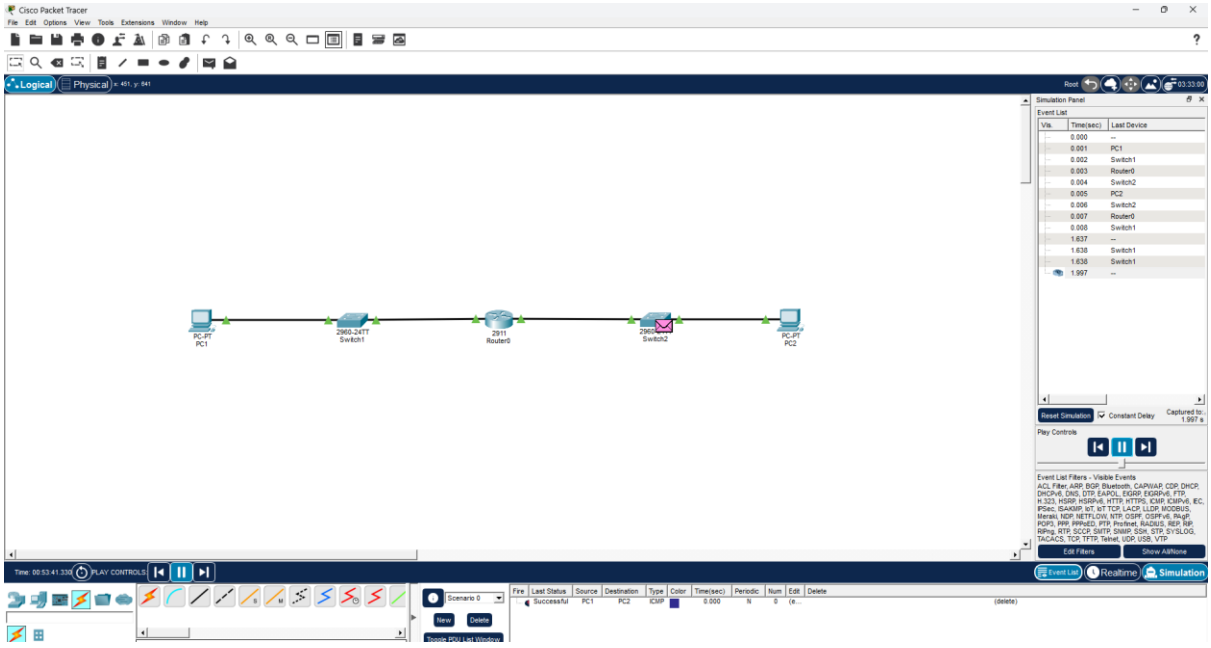
1.c

```
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0060.3e7e.de13    DYNAMIC     Fa0/1
   1    00e0.a3de.0ae3    DYNAMIC     Fa0/3
   1    00e0.f778.4888    DYNAMIC     Fa0/2
Switch#
```

2. Two departments are connected using router as shown below. After configuring the IP addresses, users report that communication between departments is unsuccessful.
   PC1 ---- Switch ---- Router ---- Switch ---- PC2
   Verify the forwarding path used by packets between the departments. Determine whether the router contains sufficient information to reach the remote network.



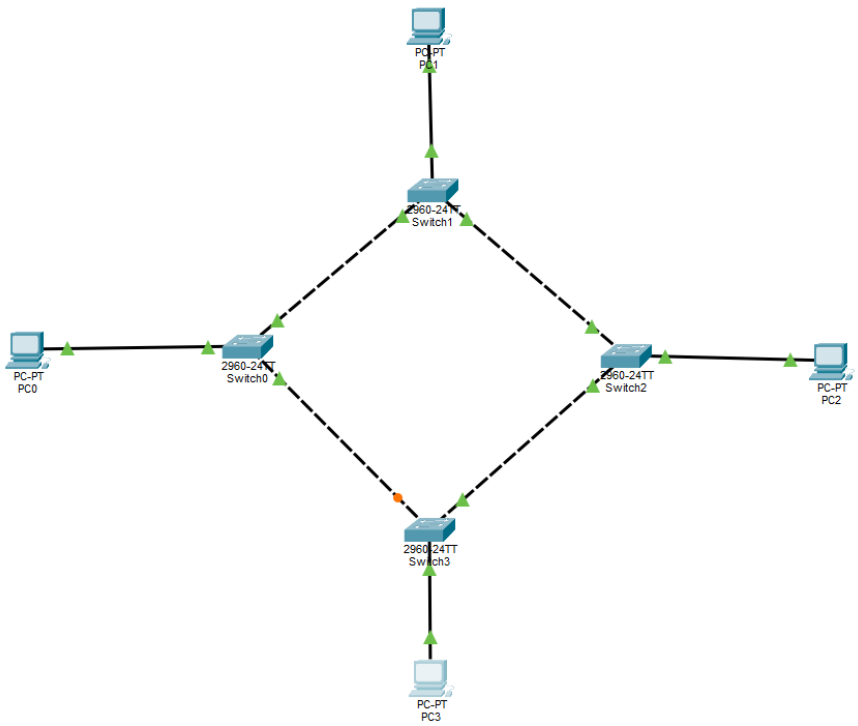
```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

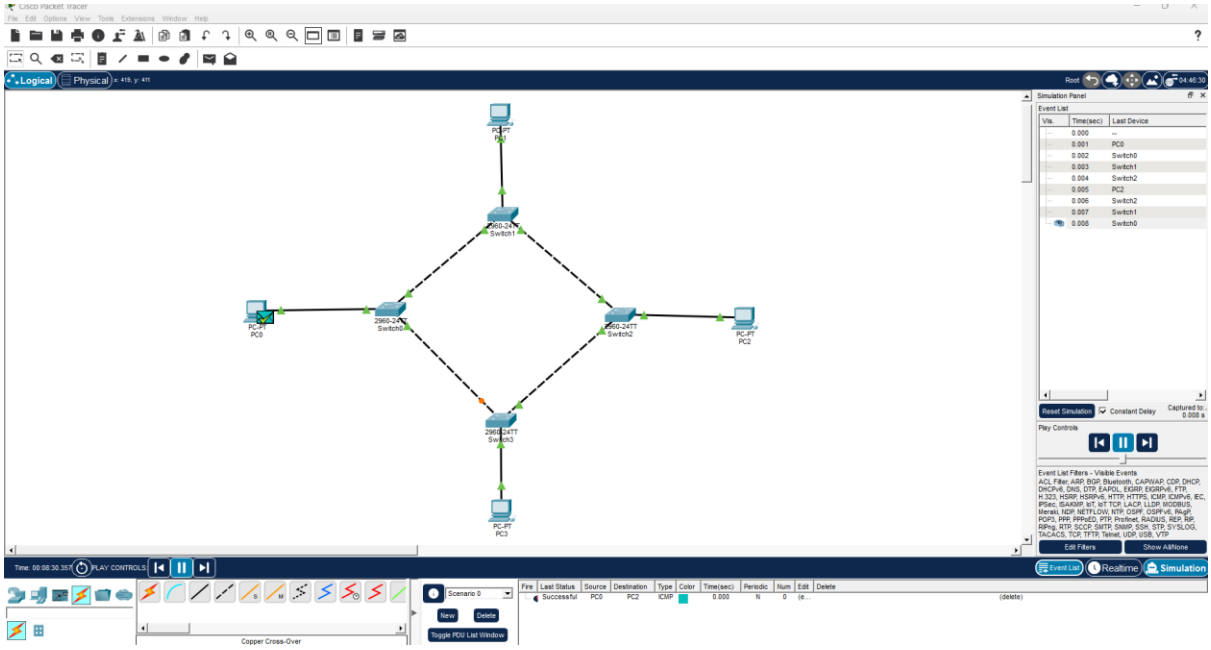
Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1

Router#
```

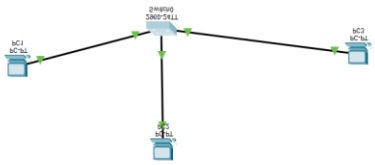
3. Design a ring topology with proper forwarding of packets.



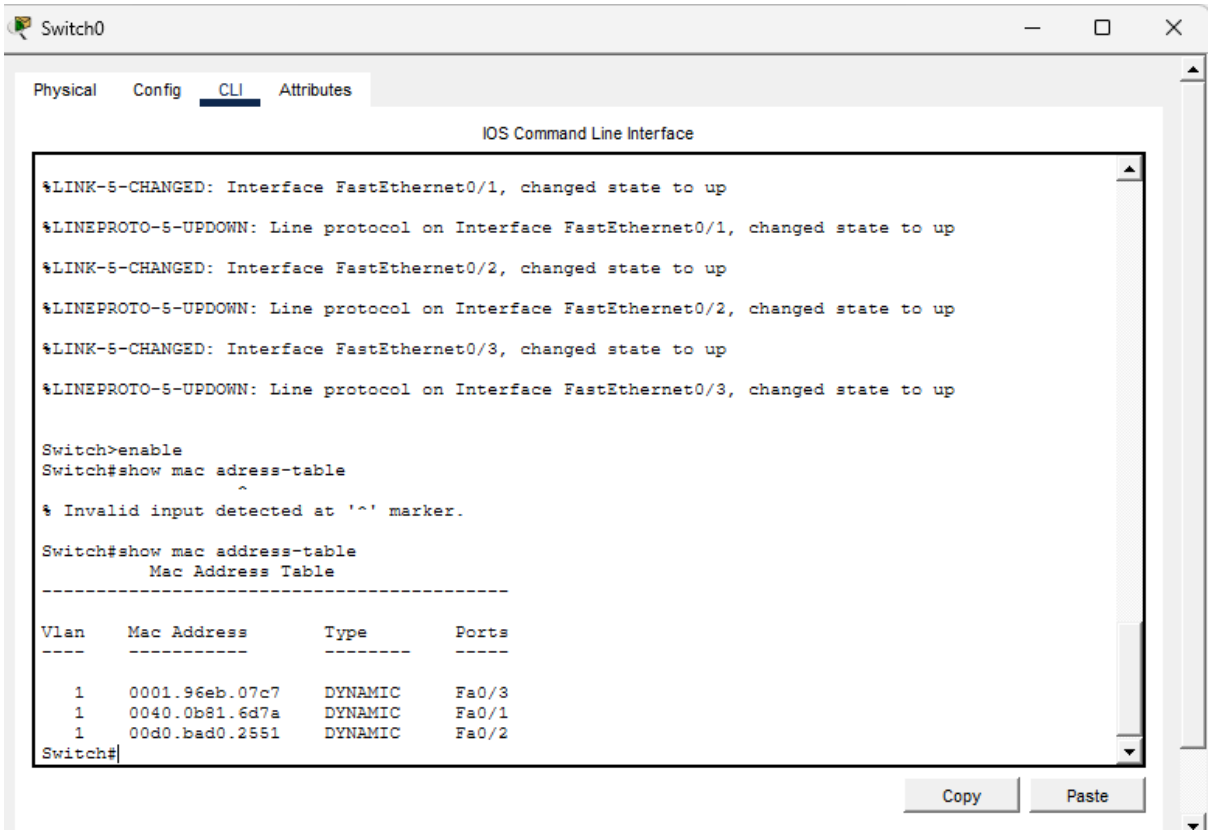


4. A small office has three computers (PC1, PC2, and PC3) connected through a switch. The network administrator wants all three computers to communicate within the same LAN.
   a. Design the above network topology in Cisco Packet Tracer.
   b. Configure the IP address and subnet mask on each PC.
   c. Verify end-to-end connectivity between all three computers using the appropriate command.
   d. Display the MAC Address Table of the switch.
   e. Display the ARP Table of PC1 after communicating with PC2.

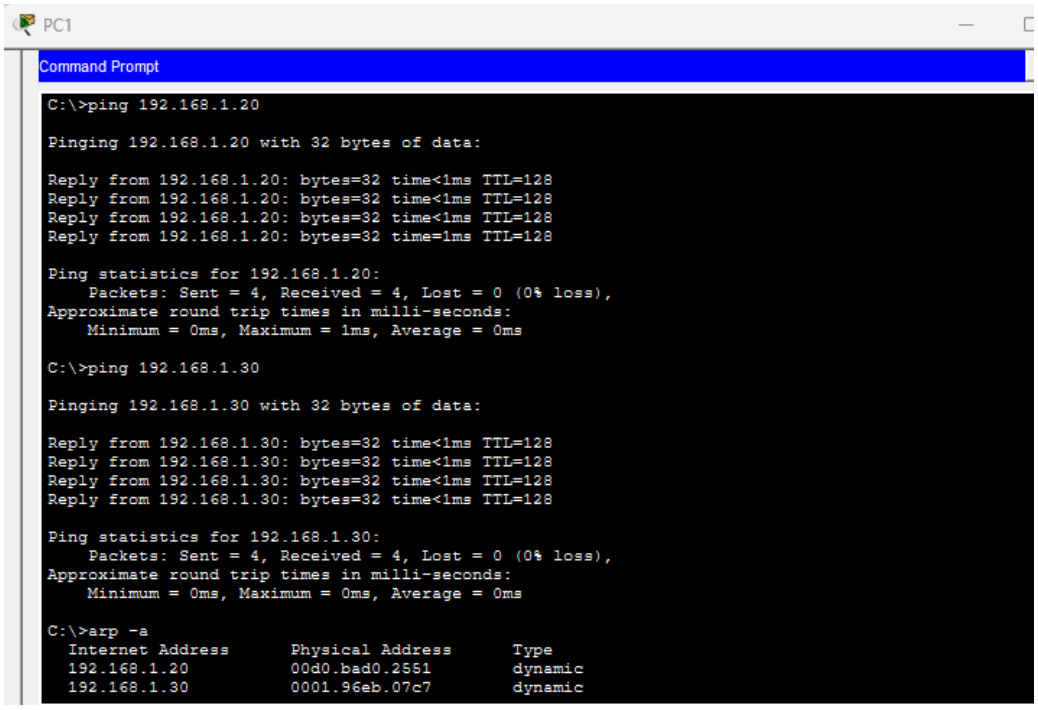
a,b.



d.


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
Switch>enable
Switch#show mac adress-table
^
% Invalid input detected at '^' marker.
Switch#show mac address-table
Mac Address Table
Vlan Mac Address Type Ports
1 0001.96eb.07c7 DYNAMIC Fa0/3
1 0040.0b81.6d7a DYNAMIC Fa0/1
1 00d0.bad0.2551 DYNAMIC Fa0/2
Switch#
Copy
Paste

c,e.


PC1
Command Prompt
C:\>ping 192.168.1.20
Pinging 192.168.1.20 with 32 bytes of data:
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.20:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.30
Pinging 192.168.1.30 with 32 bytes of data:
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.30:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>arp -a
Internet Address Physical Address Type
192.168.1.20 00d0.bad0.2551 dynamic
192.168.1.30 0001.96eb.07c7 dynamic

5. A company has four branch offices connected through four routers (R1, R2, R3 and R4). Each branch contains one LAN. The routers must communicate using Static Routing.
   a. Display the routing table of every router.
   b. Remove one static route and identify the effect.
   c. A packet transmitted from Branch A fails to reach Branch B. Determine the path followed by the packet and identify the router responsible for the communication failure.
   d. Show the network in the form of ring and check for communication failures.

a. Router1

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.41.0/30 is directly connected, GigabitEthernet0/2
L       10.0.41.2/32 is directly connected, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 [1/0] via 10.0.12.2
S    192.168.3.0/24 [1/0] via 10.0.12.2
S    192.168.4.0/24 [1/0] via 10.0.41.1
```

Router2

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.2/32 is directly connected, GigabitEthernet0/1
C       10.0.23.0/30 is directly connected, GigabitEthernet0/2
L       10.0.23.1/32 is directly connected, GigabitEthernet0/2
S    192.168.1.0/24 [1/0] via 10.0.12.1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0
S    192.168.3.0/24 [1/0] via 10.0.23.2
S    192.168.4.0/24 [1/0] via 10.0.23.2
```

Router3

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.23.0/30 is directly connected, GigabitEthernet0/1
L       10.0.23.2/32 is directly connected, GigabitEthernet0/1
C       10.0.34.0/30 is directly connected, GigabitEthernet0/2
L       10.0.34.1/32 is directly connected, GigabitEthernet0/2
S    192.168.1.0/24 [1/0] via 10.0.23.1
S    192.168.2.0/24 [1/0] via 10.0.23.1
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0
S    192.168.4.0/24 [1/0] via 10.0.34.2
```

Router4

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.34.0/30 is directly connected, GigabitEthernet0/1
L       10.0.34.2/32 is directly connected, GigabitEthernet0/1
C       10.0.41.0/30 is directly connected, GigabitEthernet0/2
L       10.0.41.1/32 is directly connected, GigabitEthernet0/2
S    192.168.1.0/24 [1/0] via 10.0.41.2
S    192.168.2.0/24 [1/0] via 10.0.34.1
S    192.168.3.0/24 [1/0] via 10.0.34.1
     192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/24 is directly connected, GigabitEthernet0/0
L       192.168.4.1/32 is directly connected, GigabitEthernet0/0
```

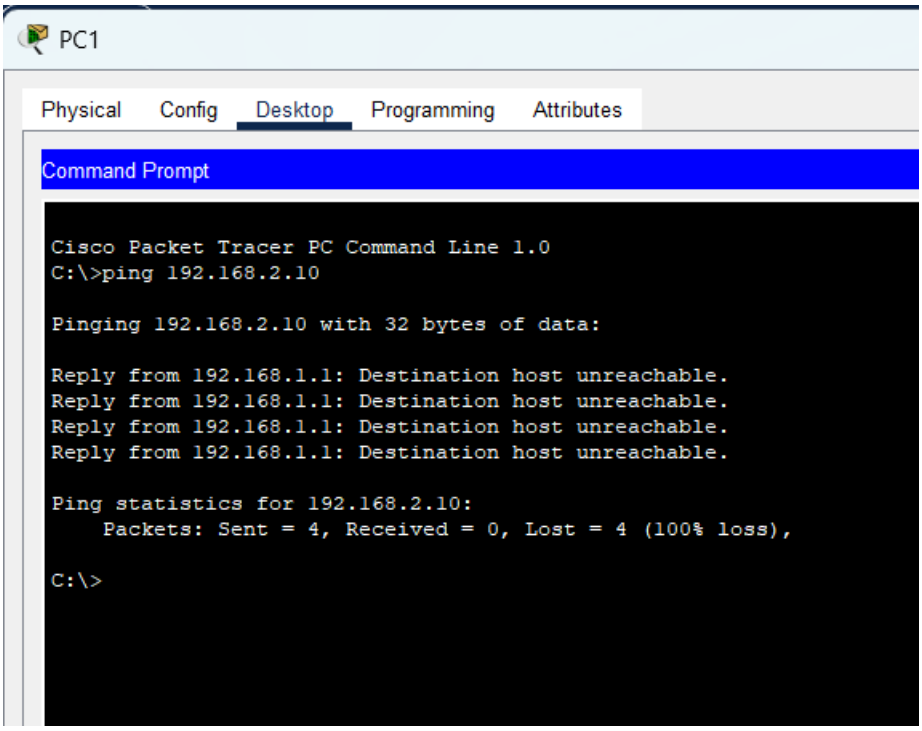
b. Router1

```
Router(config)#no ip route 192.168.2.0 255.255.255.0 10.0.12.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.41.0/30 is directly connected, GigabitEthernet0/2
L       10.0.41.2/32 is directly connected, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.3.0/24 [1/0] via 10.0.12.2
S    192.168.4.0/24 [1/0] via 10.0.41.1
```



c.

```
C:\>tracert 192.168.2.10

Tracing route to 192.168.2.10 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1
  2    0 ms      0 ms      0 ms      10.0.12.2
  3    0 ms      0 ms      0 ms      192.168.2.10

Trace complete.
```

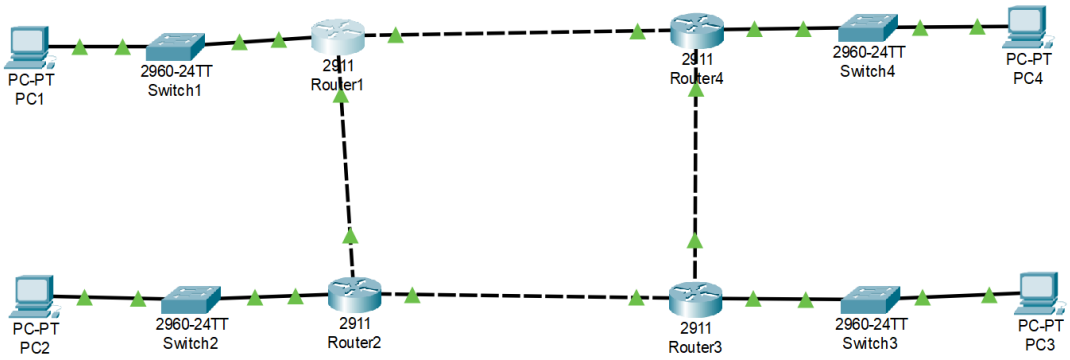
Router2

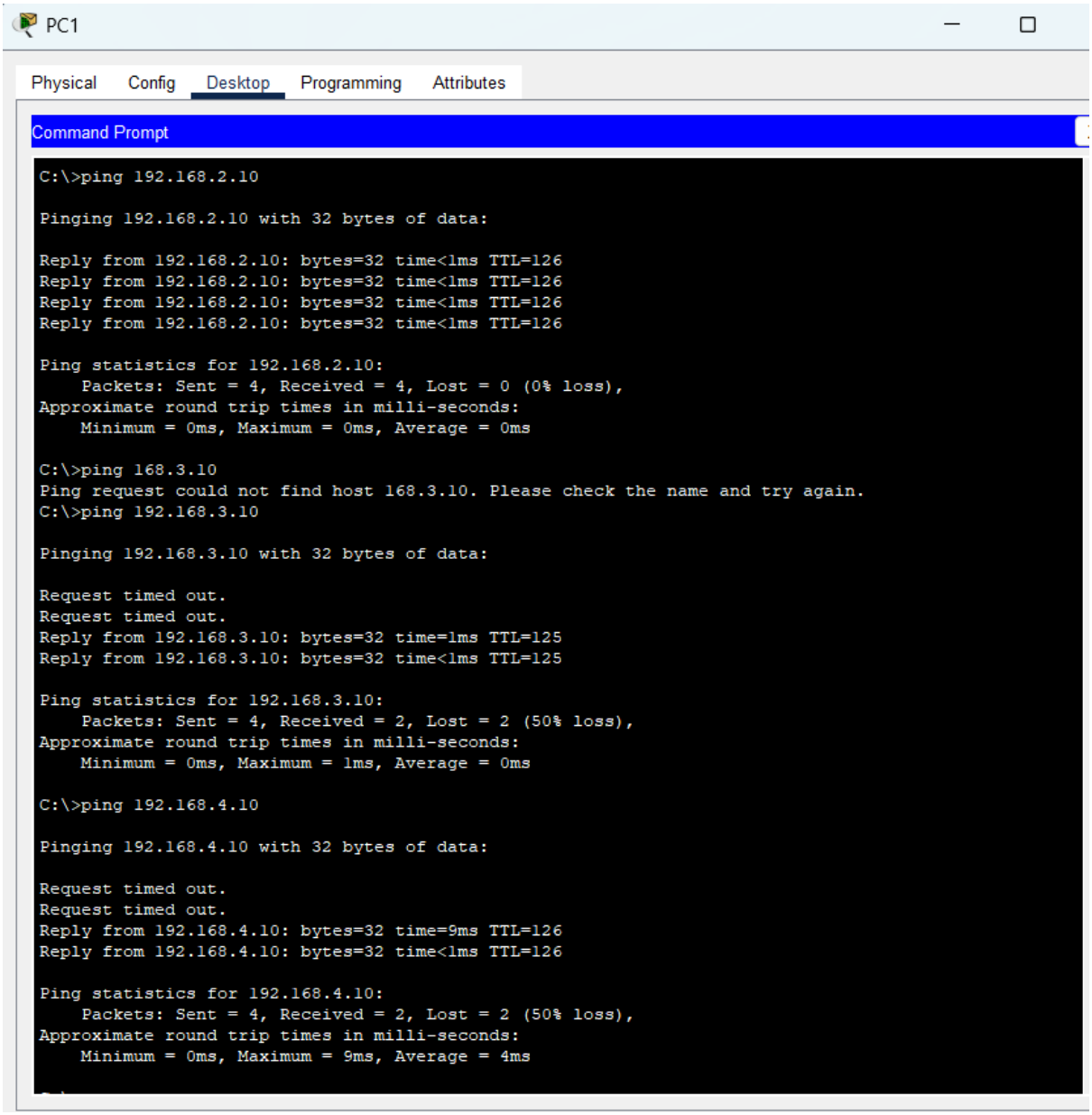
```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.2/32 is directly connected, GigabitEthernet0/1
C       10.0.23.0/30 is directly connected, GigabitEthernet0/2
L       10.0.23.1/32 is directly connected, GigabitEthernet0/2
S    192.168.1.0/24 [1/0] via 10.0.12.1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0
S    192.168.3.0/24 [1/0] via 10.0.23.2
S    192.168.4.0/24 [1/0] via 10.0.23.2
```

d.




PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 168.3.10
Ping request could not find host 168.3.10. Please check the name and try again.
C:\>ping 192.168.3.10
Pinging 192.168.3.10 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.3.10: bytes=32 time=1ms TTL=125
Reply from 192.168.3.10: bytes=32 time<1ms TTL=125
Ping statistics for 192.168.3.10:
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.4.10
Pinging 192.168.4.10 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.4.10: bytes=32 time=9ms TTL=126
Reply from 192.168.4.10: bytes=32 time<1ms TTL=126
Ping statistics for 192.168.4.10:
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 9ms, Average = 4ms